

# Touch-to-Action: Enhancing Robot Manipulation through Tactile Perception

## 1. Organizers

The Touch-to-Action workshop is jointly proposed by eight co-organizers (a Principal Investigator, an Associate Professor, a Full Professor, an Industry Leader, two Postdoctoral Researchers, and two Ph.D. candidates). Table 1 lists the details of each organizer.

- The **organizers are from five nationalities** (Italy, Germany, China, Austria, United States) and are affiliated with institutions **located in five countries** (Italy, Japan, United Kingdom, Germany, United States).
- The organizing team comprises researchers with established contributions to tactile perception, manipulation, and robot learning, aligning with the central themes of this workshop.
- **Previous workshops:** The organizers have previously served on the organizing committees of numerous workshops, with a combined experience of coordinating more than 30 events. Notable examples include: <sup>1</sup> <sup>2</sup> <sup>3</sup> <sup>4</sup> <sup>5</sup> <sup>6</sup> <sup>7</sup>.

Table 1: Co-organizers of the Touch-to-Action workshop proposal. (\*) indicates the contact person.

| Name                  | Affiliation                                  | E-mail                     | Website                                                                                                                                   | Previous Experience                                                                               |
|-----------------------|----------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Carlo Alessi (*)      | Italian Institute of Technology              | carlo.alessi@iit.it        | <a href="https://www.iit.it/it/people-details/-/people/carlo-alessi">https://www.iit.it/it/people-details/-/people/carlo-alessi</a>       | Organized 1 workshop at an IEEE robotics conference (ARM 2026)                                    |
| Gabriele Mario Caddeo | Italian Institute of Technology              | gabriele.caddeo@iit.it     | <a href="https://www.iit.it/it/people-details/-/people/gabriele-caddeo">https://www.iit.it/it/people-details/-/people/gabriele-caddeo</a> | /                                                                                                 |
| Iris Andrussow        | Max Planck Institute for Intelligent Systems | andrussow@is.mpg.de        | <a href="https://is.mpg.de/person/andrussow">https://is.mpg.de/person/andrussow</a>                                                       | Organized 1 workshop at an IEEE robotics conference (ICRA 2022)                                   |
| Yuxiang Ma            | Massachusetts Institute of Technology        | yxma20@mit.edu             | <a href="https://yuxiang-ma.github.io/">https://yuxiang-ma.github.io/</a>                                                                 | Organized 1 workshop at RSS 2025                                                                  |
| Alexander Schmitz     | XELA Robotics                                | alexander@xelarobotics.com | <a href="https://www.xelarobotics.com/about-xela">https://www.xelarobotics.com/about-xela</a>                                             | Organized 1 workshop at an IEEE robotics conference (ICRA 2017)                                   |
| Lorenzo Jamone        | University College London                    | l.jamone@ucl.ac.uk         | <a href="https://profiles.ucl.ac.uk/102308-lorenzo-jamone">https://profiles.ucl.ac.uk/102308-lorenzo-jamone</a>                           | Organized more than 20 workshops at major IEEE robotics conferences (IROS, ICRA, ICDL, Humanoids) |
| Mark Cutkosky         | Stanford University                          | cutkosky@stanford.edu      | <a href="https://me.stanford.edu/people/mark-cutkosky">https://me.stanford.edu/people/mark-cutkosky</a>                                   | /                                                                                                 |
| Lorenzo Natale        | Italian Institute of Technology              | lorenzo.natale@iit.it      | <a href="https://lornat75.github.io/index.html">https://lornat75.github.io/index.html</a>                                                 | Organized more than 10 workshops at major robotics conferences (IROS, ICRA, RSS, Humanoids, HRI)  |

<sup>1</sup>Carlo Alessi: Safety-Critical Software, Control and Learning for Robotics, **ARM 2026** (<https://sites.google.com/view/safehriicarm26/home>);

<sup>2</sup>Yuxiang Ma: 1st Workshop on Robot Hardware-Aware Intelligence, **RSS 2025** (<https://rss-hardware-intelligence.github.io/>);

<sup>3</sup>Iris Andrussow: Workshop Debates on the Future of Robotics Research, **ICRA 2022** (<https://www.roboticsdebates.org/past-debates>);

<sup>4</sup>L. Jamone: Robot Learning From Human Teleoperation: Prospects and Challenges, **IROS 2025** (<https://sites.google.com/view/lfht-iros25-workshop-proposal/home>)

<sup>5</sup>L. Natale, L. Jamone, A. Schmitz: The Robotic Sense of Touch, **ICRA 2017** (<https://roboticsenseoftouchws.wordpress.com/>);

<sup>6</sup>L. Natale, L. Jamone: Autonomous Object Manipulation, **IROS 2019** (<https://manipulation-iros-workshop.github.io/>)

<sup>7</sup>L. Natale, L. Jamone: Human-Robot Handovers, **ICRA 2020** (<https://handovers.gitlab.io/icra2020-handovers/>);

## 2. Abstract (max 250 words) and Website

**Abstract:** Tactile sensing is fundamental for humans to perceive the world and manipulate objects. In robotics, Artificial Intelligence (AI) is transforming manipulation by leveraging proprioception, vision, and language. However, touch in manipulation remains underexplored, with research still fragmented compared to other sensing modalities (e.g., vision), which benefit from standardized technologies. **The Touch-to-Action workshop provides a roadmap to tactile sensing and manipulation** with a vertical approach, bridging bioinspiration, sensor design and fabrication, simulation, perception, and control. First, we review the biological sense of touch and how it guides the development of robotic tactile sensors, setting the fundamentals for subsequent technical sessions. Then, we cover tactile sensor technologies, including transduction principles, design and fabrication methods, and integration strategies. We also address the promising role of simulation, from supporting optimal sensor design to facilitating policy learning. Moreover, we explore deep learning methods for tactile perception and multimodal sensor fusion, which leads us to discuss how tactile feedback is used in robot control for manipulation. Applications in manufacturing and healthcare will be highlighted with examples from industry and research. This half-day multi-disciplinary event gathers diverse robotics and AI communities, offering interactive **Keynote Presentations**, an **Industry Session**, a **Panel Discussion**, **Lightning Talks**, and **Poster Sessions**.

**Website:** The workshop website (under construction) is reachable here:  
<https://iros-touch2action.github.io/>.

## 3. Content of the workshop and expected impact

### Objectives of the workshop:

The Touch-to-Action workshop aims to provide a comprehensive roadmap of tactile sensing and its vertical integration throughout the modern robotic learning and manipulation pipeline. To achieve this goal, we define the following strategic objectives:

1. Establish a Holistic Multidisciplinary Framework: We aim to equip the community with a unified understanding of the entire tactile manipulation chain, spanning bioinspiration, sensor design and fabrication, high-fidelity simulation, perception, and learning-based control.
2. Foster Community Cross-Pollination: To bridge the divide between researchers focused on hardware/materials and those focused on AI/control. By creating a shared roadmap, we aim to help newcomers find entry points into the field while offering experienced researchers a platform for interdisciplinary collaboration.
3. Align Academic Innovation and Industrial Requirements: We facilitate a two-way dialogue between academia and industry stakeholders to identify deployment challenges, scalability requirements, and emerging research directions in tactile sensing. This exchange also highlights commercial technologies that can support and accelerate academic research.

### Does the content reflect the state of the art in a subfield of robotics and automation?

The content of the Touch-to-Action workshop represents the state of the art in the following fields: (i) tactile sensor design and fabrication, (ii) deep learning for tactile perception, and (iii) robot learning for dexterous manipulation. The invited keynote speakers are internationally recognized pioneers who are defining the frontier of Robotics & AI. Their contributions consistently appear in top-tier robotics venues.

The inclusion of an Industry Session further ensures that the content discussed is not only intellectually stimulating but also applied in real-world applications.

### How will you engage the attendees and exchange ideas?

We aim to create a dynamic and interactive forum that goes beyond a standard sequence of talks. To ensure active participation and knowledge exchange, we will implement the following strategies:

- **Distributed Q&A (during Keynotes):** To maintain high cognitive engagement, we will move beyond the traditional end-of-talk Q&A format. Speakers will be encouraged to pause at logical transitions for brief questions, ensuring the audience remains synchronized with complex technical material.
- **Organizer-participant interaction:** Before the workshop, we will organize *online seminars* where participants can introduce their research and interact with the other attendees. This early exchange will help participants become familiar with each other’s work and identify shared interests. During the poster session, the organizers will actively circulate among presenters and attendees to promote deeper discussions and facilitate mentorship and networking opportunities. We will also encourage video and hardware demonstrations, fostering more concrete and interactive exchanges around ongoing research.
- **Interactive Panel Discussion:** The panel will combine prepared questions on key research challenges with questions raised by the audience. To use time efficiently, we will leverage *Slido*, a digital Q&A and polling platform, to submit and upvote questions in real-time. This approach will help ensure that the discussion focuses on the hottest topics.

### How will the proposed event expand the originality and diversity of workshop content at IROS 2026?

The proponents believe that a holistic treatment of tactile sensing and manipulation is both timely and novel. While vision-language-action (VLA) models have significantly advanced robotic manipulation, tactile sensing has the potential to unlock new capabilities in contact-rich and dexterous tasks where touch provides essential information. Advancing tactile sensing and manipulation requires a multidisciplinary perspective spanning sensor design, simulation, perception, and robot learning. However, these topics are often discussed in isolation. The *Touch-to-Action* workshop aims to address this gap by providing a unified overview of the role of tactile sensing across the full robotic manipulation pipeline. A key contribution of the workshop is to bridge communities that typically interact only partially, including researchers in robotics, machine learning, sensor design, and biomechanics. By bringing together these perspectives, the workshop aims to identify current limitations and highlight promising directions for future research. The interdisciplinary topics of the *Touch-to-Action* workshop are structured in five interconnected sessions, described as follows.

- **Session I: The Sense of Touch — From Humans to Robots.** The introductory session reviews the fundamentals of the sense of touch in humans and other biological systems. The aim is to answer:

*How does biological touch work, and how does it inspire artificial tactile systems?*

The session highlights the fundamental mechanisms of biological touch and discusses how they guide sensor development, placement, and multimodal integration in robotic systems. The aim is to provide participants with the biological foundations of touch, paving the way for the remaining technical sessions.

- **Session II: Development & Integration of Tactile Sensors.** This session explores the development of tactile sensors, covering transduction technologies, design and fabrication, and system integration strategies. We review core transduction methods that convert physical contact into measurable signals (e.g., magnetic, optical), raising the practical question:

*Which transduction method is most appropriate for a given application?*

The choice of transduction technology often enables or constrains sensor morphology. We therefore discuss the design of tactile sensors, including their overall shape (e.g., flat, curved) and component-specific structures (e.g., artificial skin, fingertips). The session also examines fabrication techniques, such as molding of soft materials and additive manufacturing. Finally, we address best practices for integrating tactile sensors into robotic platforms while balancing mechanical robustness and sensing transparency. Examples from research and industry will highlight key challenges and design-performance trade-offs.

- **Session III: Simulation of Tactile Sensors.** This session examines the promising role of simulation in the development and application of tactile sensors. On the one hand, high-fidelity finite element methods (e.g., SOFA) can precisely model complex geometries, contact mechanics, and deformation of soft materials, supporting iterative sensor design and accurate performance prediction. On the other hand, real-time rigid-body simulators (e.g., MuJoCo) offer fast environments for embedding tactile feedback into control policies. Despite the effectiveness of these simulations, the *sim-to-real* gap persists due to simplifying assumptions and unmodeled dynamics. This raises a central question:

*How can we achieve effective tactile simulations? What is missing?*

The session examines the trade-offs between accuracy and computational efficiency. Topics include best practices for tactile data collection, data-driven optimization to align simulated and physical sensor data, and hybrid modeling techniques to improve simulation fidelity.

- **Session IV: Tactile Perception & Processing.** This session explores how to process tactile data to extract contact information that is suitable for high-level perception and control purposes. Modern tactile sensors can perceive several fundamental contact information, including contact forces, surface texture, and slip. In this regard, both the sensor embodiment (e.g., shape, materials, and physical placement) and the task performed influence perception and the interpretation of the information. Tactile data can take various forms and dimensionalities, raising the question:

*How should tactile data be processed and fused with other sensing modalities?*

We cover deep learning methods applied to high-dimensional tactile data in perception tasks. Topics include domain adaptation to bridge inter-sensor discrepancies or the *sim-to-real* gap, generative models for data augmentation, and multimodal sensor fusion with complementary sensing modalities.

- **Session V: Tactile Sensing in Robot Learning and Manipulation.** The final session focuses on how tactile sensing can be leveraged in multimodal control policies to enable dexterous and adaptive manipulation. We want to answer the question:

*When is touch not just useful, but essential for manipulation?*

Emphasis will be placed on imitation learning methods, *sim-to-real* transfer of reinforcement learning policies, and on leveraging human data for robotic manipulation. We discuss current

limitations, open challenges, and promising solutions to improve generalization and reliability in real-world deployment. Beyond control, the workshop examines the impact of tactile manipulation across key application sectors (e.g., manufacturing, agriculture, healthcare), identifying scenarios where tactile sensing provides the highest advantages.

The workshop will benefit both newcomers and experienced researchers interested in tactile manipulation. Participants working on specific aspects of the problem will have the opportunity to interact with experts from complementary areas, fostering cross-disciplinary discussion and new collaborations. The workshop also welcomes diverse viewpoints. Researchers who believe that touch will play a limited role in future robotic systems will be invited to share their perspectives during the Interactive Panel Discussion, encouraging constructive debate within the community. Finally, an industry session will help bridge the gap between academic research and real-world applications. By facilitating dialogue between researchers and industry stakeholders, the workshop aims to promote the translation of advances in tactile sensing into deployable robotic systems. We believe that this interdisciplinary and community-driven format will contribute to expanding the originality and diversity of workshops at IROS 2026.

#### 4. Intended audience and expected attendance

**Intended audience:** The Touch-to-Action workshop targets students, researchers, and practitioners from academia and industrial R&D working on robotics, AI, and sensor technologies. The primary audience includes researchers with backgrounds in tactile sensing and perception, robotic grasping and manipulation, robot learning, and machine learning for robotics. In addition, the workshop is designed to be accessible to researchers who are not specialists in tactile sensing but are interested in integrating touch into their robotic systems. By presenting a vertical overview of the full tactile manipulation pipeline, the workshop provides a roadmap for newcomers entering the field.

**Targeted RAS Technical Committees (TCs):** The most relevant communities include the RAS TC on *Robotic Hands, Grasping and Manipulation* (RAS-RHGM), *Robot Learning* (RAS-RL), and *Haptics* (RAS-H). Chairs and members of these TCs will be invited to join the workshop and interact with the participants.

**Expected Participation:** Given our past experience, we anticipate an audience between 100 and 150 participants, with 70-100 attending in person and the remainder joining virtually, depending on conference logistics.

#### 5. Invited speakers

**Keynote Speakers:** Table 2 lists the five keynote speakers and panelists, along with their affiliations, tentative presentation titles, and inclusion criteria. All speakers are world-leading researchers in robotics and AI (tactile perception, manipulation), with mixed experience working in academia and industry. The speakers come from three different continents (North America, Europe, Asia) and are affiliated with institutions across two continents (North America and Europe). Their career stages range from Assistant Professor to Full Professor. Forty percent (40%) of the invited speakers are women, favoring gender diversity among participants. All speakers confirmed their participation in person. Confirmation emails from each speaker are provided in the Appendix.

**Industry Representatives:** Beyond the Keynote speakers, we are also inviting industry representatives to connect academia and industry in the field of tactile perception and manipulation. Table 3 reports three representatives of innovative companies and one industry expert involved

Table 2: List of invited speakers for the Keynote Presentations.

| Invited Speaker                | Affiliation                             | Confirmed | Tentative title                                                                                                     | Inclusion Criteria                                                                                 |
|--------------------------------|-----------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Allison Okamura (Full Prof.)   | Stanford University                     | Yes       | Wearable Pneumatics for Bidirectional Tactile Interaction: From Human Communication to Training Autonomous Policies | Leading researcher in haptics; Senior female researcher.                                           |
| Wenzhen Yuan (Assistant Prof.) | University of Illinois Urbana-Champaign | Yes       | Towards Generalizable Tactile Sensors and Sensing via Simulation                                                    | Leading expert in vision-based tactile sensors development (GelSight); Female researcher.          |
| Nathan Lepora (Full Prof.)     | University of Bristol                   | Yes       | Sim-to-real tactile sensing for robot control                                                                       | Leading researcher in tactile manipulation and biomimetic sensing; Strong academia-industry links. |
| Nima Fazeli (Assistant Prof.)  | University of Michigan                  | Yes       | Scaling Tactile Perception and Learning: Sim2Real Transfer and Learning from Humans                                 | Expert in tactile perception, sim-to-real, and contact-rich manipulation.                          |
| Lerrel Pinto (Assistant Prof.) | New York University                     | Yes       | Robot Learning from Human Touch                                                                                     | Expert in robot learning; Academia-industry experience.                                            |

in the robotics and AI sector, particularly for designing and manufacturing tactile sensors and dexterous robotic hands. These companies are headquartered in three different countries (Japan, China, and USA), reflecting a wide geographical representation. Confirmation emails are provided in the Appendix.

Table 3: List of confirmed representatives of the Industry Session.

| Invited Speaker                         | Company                                                                                                    | Sponsors Award         | Inclusion Criteria                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------|
| Alexander Schmitz (CEO)                 | XELA Robotics, Japan ( <a href="https://www.xelarobotics.com/">https://www.xelarobotics.com/</a> )         | ✓(300 EUR)             | Academia-industry profile; Company developing magnetic tactile sensors widely adopted in robotics research. |
| Matei Ciocarlie (Chief Science Officer) | Tangent Robotics AI, USA ( <a href="https://www.tangentrobotics.ai/">https://www.tangentrobotics.ai/</a> ) | ✗                      | Academia-industry profile; Start-up developing dexterous robotic hands.                                     |
| Jianghua Duan (CEO)                     | Daimon Robotics, China ( <a href="https://www.dmrobot.com/en/">https://www.dmrobot.com/en/</a> )           | ✓ (Registration grant) | Industry profile; Company develops tactile sensors, hardware and software for manipulation.                 |
| Jeremy Fishel (Principal Scientist)     | Former Sanctuary AI, now independent expert                                                                | ✗                      | Academia-industry profile; Expert in tactile sensor development (BioTac) and perception.                    |

## 6. Structure of the workshop

Touch-to-Action is structured to guide participants through the full tactile manipulation pipeline while creating multiple opportunities for discussion and interaction. The workshop combines keynote presentations, contributed talks, posters, industry demonstrations, and a panel discussion. In addition, pre-workshop online events and post-workshop collaboration activities are planned to extend interactions beyond the in-person meeting.

**Keynote Presentations with Distributed Q&A:** Leading experts in tactile sensing and manipulation will deliver keynote presentations covering topics such as the physical fundamentals of touch sensing, sensor design and fabrication, simulation, tactile perception, and robot learning. Each speaker will present recent advances, discuss the role of touch in robotic manipulation, and

outline future research directions. To encourage interaction, each keynote will include a *Distributed Q&A* format. Participants will be able to ask short clarification questions at intermediate checkpoints during the talk, as well as at the end of the presentation. Speakers will also be encouraged to actively engage with the audience throughout their talk.

**Industry Session with Live Demonstrations:** Representatives from four companies will present their latest technologies and perspectives on the state of the art in tactile sensing and robotic manipulation. Whenever possible, presentations will be complemented by short live demonstrations. These talks will highlight real-world deployment challenges and showcase solutions that are currently being developed or commercialized. The session will provide researchers with insight into how commercial technologies can support their work, while allowing companies to gather feedback on emerging research needs. We will encourage the company representatives to interact with the participants.

**Interactive Panel Discussion:** To encourage active discussion between established experts and early-career researchers, the workshop will include a 35-minute interactive panel discussion with invited speakers, including Prof. Mark Cutkosky. Questions and discussion points from the audience will be actively encouraged. In addition, the author of the best submitted mini-review will join the panel, bringing an informed early-career perspective from the community. To further broaden participation, audience members who wish to challenge or complement a panelist’s viewpoint may be invited to join the panel discussion. Any remaining buffer time in the schedule will be allocated to this session to allow for extended discussion.

**Paper Contributions, Lightning Talks, and Poster Session:** To highlight contributions from early-career researchers, we will issue a call for contributions in the form of short papers and mini-reviews. The top submitted research papers will be selected for short lightning talks, giving young researchers visibility for their work. These contributions, together with the other accepted papers, will also be presented as posters during the breaks, creating additional opportunities for discussion and feedback. We encourage two types of submissions, which will be reviewed transparently through **OpenReview**:

- **Short Papers** (3–4 pages) that present ongoing work, preliminary results, or novel ideas related to the workshop themes. The goal is to encourage early dissemination, foster collaboration, and give visibility to promising research that may not yet be ready for full conference submission. Authors will be encouraged to bring prototype demonstrations, and selected papers may be invited to participate in lightning talks in addition to poster presentations.
- **Mini-Reviews and Perspectives** (2–4 pages) that summarize recent advances, identify open challenges, and highlight emerging research directions in tactile sensing and manipulation. Outstanding contributions will help inform the interactive panel discussion. The author of the best mini-review will be invited to join the panel, representing the perspective of an informed early-career researcher. Furthermore, we intend to collaborate with interested authors to develop a joint review paper for an IEEE journal.

**Pre-workshop Online Seminars:** To foster early networking, we plan to host up to four online seminars in the month preceding the workshop (approximately one per week). These sessions will allow participants who submitted papers to present their work and research interests in advance. By becoming familiar with each other’s research before the in-person meeting, participants can identify shared interests, establish connections, and enable deeper discussions during the workshop.

**Post-workshop Collaboration:** Following the workshop, we aim to collaborate with authors of the submitted mini-reviews to develop a joint perspective paper summarizing recent advances and open challenges in tactile sensing and manipulation.

## 7. Tentative schedule

Table 4 shows the planned workshop schedule<sup>8</sup>. Each keynote speaker is allocated a 20-minute slot (15 minutes for the keynote presentation and 5 minutes for Q&A).

The Industry Session is scheduled for 35 minutes and will include brief 8-minute presentations from each industry representative. The interactive panel discussion will last 30 minutes. Early-career researchers will be highlighted during a 10-minute Lightning Talk Session and through poster presentations held during the coffee breaks.

Buffer time is included throughout the schedule to accommodate transitions, potential delays, and extended discussion.

Table 4: Planned schedule of the half-day workshop.

| Time                   | Talk                                                                                                                                                   | Comments                                                         |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 8:30 - 8:40<br>(5-10') | Introduction                                                                                                                                           | Includes buffer time to handle transitions and unforeseen delays |
| 8:40 - 9:00<br>(20')   | Prof. <b>Allison Okamura</b><br>Wearable Pneumatics for Bidirectional Tactile Interaction:<br>From Human Communication to Training Autonomous Policies | Session 1<br>(KP + Distributed Q&A)                              |
| 9:00 - 9:20<br>(20')   | Prof. <b>Whenzhen Yuan</b><br>Towards Generalizable Tactile Sensors<br>and Sensing via Simulation                                                      | Session 2<br>(KP + Distributed Q&A)                              |
| 9:20 - 9:40<br>(20')   | Prof. <b>Nathan Lepora</b><br>Sim-to-real tactile sensing<br>for robot control                                                                         | Session 3<br>(KP + Distributed Q&A)                              |
| 9:40 - 10:00<br>(20')  | Prof. <b>Nima Fazeli</b><br>Scaling Tactile Perception and Learning: Sim2Real Transfer<br>and Learning from Humans                                     | Session 4<br>(KP + Distributed Q&A)                              |
| 10:00 - 10:20<br>(20') | Prof. <b>Lerrel Pinto</b><br>Robot Learning from Human Touch                                                                                           | Session 5<br>(KP + Distributed Q&A)                              |
| 10:20 - 10:30<br>(10') | <b>Lightning Presentations</b>                                                                                                                         | (1' each)                                                        |
| 10:30 - 11:00<br>(30') | Coffee Break & <b>Interactive Poster Session</b>                                                                                                       |                                                                  |
| 11:00 - 11:35<br>(35') | <b>Industry Session</b>                                                                                                                                | Presentations + Q&A (8' each)                                    |
| 11:35 - 12:10<br>(35') | <b>Interactive Panel Discussion</b>                                                                                                                    | All Keynote Speakers<br>& Prof. Mark Cutkosky                    |
| 12:10 - 12:25<br>(15') | Buffer time                                                                                                                                            | Allows to handle transitions<br>and unforeseen delays            |
| 12:25 - 12:30<br>(5')  | <b>Awards to Best Lightning Talks</b>                                                                                                                  | Sponsored by XELA Robotics                                       |
| 12:30                  | End                                                                                                                                                    |                                                                  |

<sup>8</sup>KP = Keynote Presentation, Q&A = Question and Answer



## 8. Plan to solicit participation

We will encourage broad participation and inclusion from the wider robotics community beyond the invited speakers by actively promoting the workshop in multiple ways.

- **Mailing lists:** We will advertise the event through academic mailing lists (e.g., Robotics Worldwide, Reinforcement Learning Mailing List, IEEE RAS TC contact points).
- **Social Media:** We will advertise the event through our personal and institutional social media platforms (e.g., LinkedIn). To further boost visibility, we will prepare short, engaging interviews with the invited speakers, both to advertise the workshop and to introduce the speakers to the community.
- **Social Event and Seminars:** Before the workshop, we will host online seminars to encourage early interaction among participants. In addition, we will organize an informal ice-breaker social event the evening before the workshop, allowing attendees to connect and start building relationships ahead of the technical sessions.
- **Call for Submission and Peer-review:** We will advertise a call for high-quality submissions from a diverse pool of participants. To this end, we will leverage our global network to contact group leaders across career stages, genders, ethnicities, and geographic regions. We aim to fairly represent diverse perspectives on tactile sensing and manipulation. To further increase participation, the organizers will invite external reviewers through OpenReview.
- **Merit-based Awards:** Accepted works will be presented as Lightning Talks or Posters, providing visibility to emerging and established researchers. In our effort to encourage participation and recognize excellence, we will award one **Best Lightning Talk** prize (sponsored by **Xela Robotics**). Candidates will be evaluated democratically with a score computed as follows: 40% from the organizers, 20% from industry representatives, and 40% from real-time audience voting following the Lightning Talks. This transparent and community-driven process ensures balanced recognition of both scientific quality and audience impact. In addition, the author of the best Mini-Review will have the opportunity to join the Interactive Panel Discussion, offering them a prominent role in shaping the debate.
- **Inclusiveness Plan (Registration grant and mentorship):** We are committed to fostering a diverse environment by extending outreach to underrepresented communities, such as Women in Robotics and Black in Robotics. To improve accessibility, we are engaging with community founders to facilitate travel grants for underrepresented researchers and students. During the workshop, we will also offer mentorship and networking opportunities for participants from these communities. In addition, **Daimon Robotics** will support the workshop by providing a registration grant. To further lower barriers to participation, we accept Short Papers (3–4 pages) of preliminary work and Mini Reviews (2–4 pages) summarizing recent advances and open challenges.

**9. Dissemination** To maximize outreach and long-term impact, we will implement a dissemination strategy that ensures accessibility of workshop materials and promotes continued collaboration within the community.

**Special Issue and Research Dissemination:** A long-term outcome of the workshop will be the preparation of a Special Issue dedicated to tactile sensing and robot learning for manipulation.

The organizers will contact the editorial boards of relevant robotics journals (e.g., *IEEE Transactions on Robotics (T-RO)*, *IEEE Robotics and Automation Letters (RA-L)*, and *IEEE Robotics & Automation Magazine (RAM)*) to explore the possibility of hosting this Special Issue. Authors of selected workshop contributions will be invited to submit extended versions of their work, which will undergo the standard peer-review process of the target venue. In addition, the submitted *Mini-Reviews and Perspectives* will be further developed and integrated into a community-driven perspective article summarizing recent advances, open challenges, and future research directions in tactile manipulation. The organizers will coordinate with the contributing authors to prepare and submit this manuscript to the Special Issue or to a relevant robotics journal (e.g., *IEEE Transactions on Robotics*).

**Archiving and Online Availability:** The workshop website will host slides and recordings from the keynote talks, as well as all workshop materials, including contributed papers, mini-reviews, posters, panel discussion recordings, and industry demonstrations. When possible, we will also share aggregated responses to polls, questions, and suggestions collected via *Slido* during the panel discussion. Accepted contributions will remain available on OpenReview and will be organized by topic on the workshop website to facilitate discovery. After the workshop, we will promote the website and its materials through social media channels, such as LinkedIn.

**Community Outreach and Continued Engagement:** To broaden participation and highlight diverse perspectives within the robotics community, we will collaborate with managers of several communities (e.g., Women in Robotics, Black in Robotics). These channels will be used to share workshop highlights and promote opportunities such as the travel grant aimed at supporting participants from underrepresented groups. To encourage continued interaction among participants, we will establish a Slack channel linked to the workshop website. Finally, we will curate a “Getting Started with Tactile Manipulation” resource page (e.g., an open GitHub repository) aggregating relevant open-source libraries, datasets, hardware documentation, and summer schools to support newcomers entering the field.

**10. Equipment** To facilitate the session, we would request the following equipment:

- **Audio:** Two handheld wireless microphones for interactive Q&A sessions.
- **Seating:** A minimum of six chairs to accommodate the invited panelists on stage.
- **Exhibition:** Between 10 and 20 poster boards and five small tables for prototype demonstrations and research video displays.

# Appendix - Confirmation emails from invited speakers and panelists

Figure 1: Prof. Mark Cutkosky confirmation email.

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**Re: Invitation to workshop on tactile sensing and manipulation (IROS 2026)**

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**Da** Mark R Cutkosky <cutkosky@stanford.edu>

**Data** gio 1/29/2026 7:38 PM

**A** Carlo Alessi <carlo.alessi@iit.it>

Ciao Carlo,

I would be happy to join!

Mark C.

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Figure 2: Prof. Allison Okamura confirmation email.

Hi Iris,

Sounds good, here is a tentative title: Wearable Pneumatics for Bidirectional Tactile Interaction: From Human Communication to Training Autonomous Policies

Best,  
Allison

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On Feb 17, 2026, at 5:56 AM, Iris Andrussov <[andrussov@is.mpg.de](mailto:andrussov@is.mpg.de)> wrote:

Hello Allison,

Thank you very much for getting back to me. We are very excited to have you as a speaker. We will mention the preference for the first day in the workshop proposal, and in case the workshop is accepted but scheduled for the last day, we will find a different solution.  
For your information, we are planning with 20 min per speaker, including a short Q&A at the end of each talk. Furthermore, we will conclude the workshop with an engaging 45-minute panel discussion on tactile sensing, hoping to bring together all invited speakers for an interactive exchange of perspectives.

If possible, we would like to kindly ask for a tentative talk title for the proposal as well as a picture of you that we could put on the workshop website.  
Thank you already.

Best regards,  
Iris Andrussov

February 3, 2026 8:02 PM, "Allison Mariko Okamura" <[aokamura@stanford.edu](mailto:aokamura@stanford.edu)> wrote:

Hi Iris,  
Thanks for the invitation, I'd be happy to speak at the workshop. One caveat is that if the workshop is scheduled for the last day, I would probably have a conflict with RAS administrative meetings (in which case, perhaps my postdoc could present instead). It's fine to list me as a speaker for the purposes of your workshop proposal submission.  
Best,  
Allison

Figure 3: Prof. Whenzhen Yuan confirmation email.

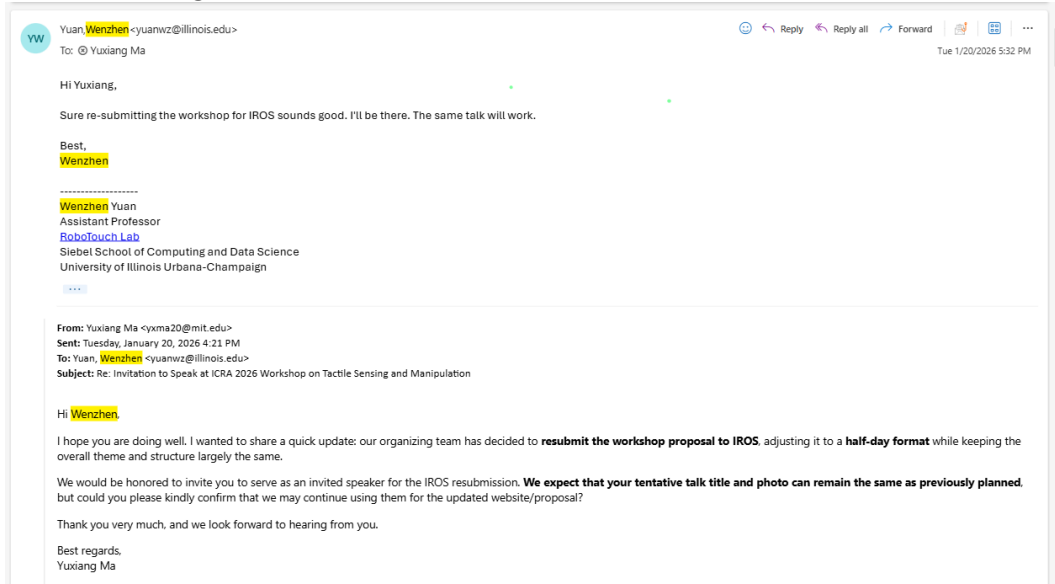


Figure 4: Prof. Nathan Lepora confirmation email.

#### Re: Invitation as Keynote speaker at IROS 2026 workshop on Tactile Sensing and Manipulation

Da Nathan Lepora <n.lepora@bristol.ac.uk>  
Data mar 2/10/2026 12:20 PM  
A Carlo Alessi <carlo.alessi@iit.it>  
Cc Benjamin Ward-Cherrier <b.ward-cherrier@bristol.ac.uk>

Hi Carlo

Thanks for the invitation - you can include me on your WS proposal, which sounds very interesting. I haven't completely made up my mind yet about attending IROS, but should be able to confirm around the time WS results are out.

thanks  
Nathan

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Nathan Lepora, Professor of Robotics & AI  
School of Engineering Mathematics & Technology, University of Bristol  
Bristol Robotics Laboratory (BRL), University of Bristol  
Office (ALB4.8): +44 117 455 1140 Lab (BRL bays 1-3)  
[www.lepora.com](http://www.lepora.com) [www.linkedin.com/in/nathan-lepora](https://www.linkedin.com/in/nathan-lepora)  
[www.bristolroboticslab.com/dexterous-robotics](http://www.bristolroboticslab.com/dexterous-robotics)

Figure 5: Prof. Nima Fazeli confirmation email.

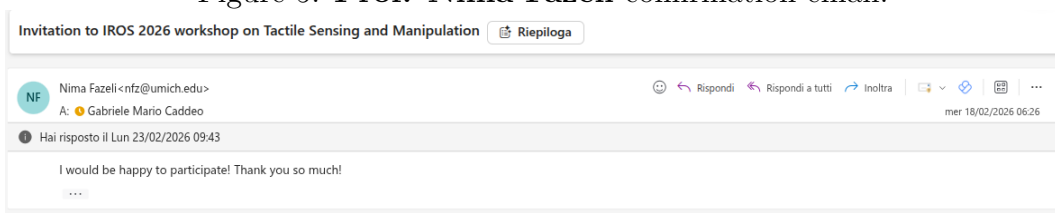
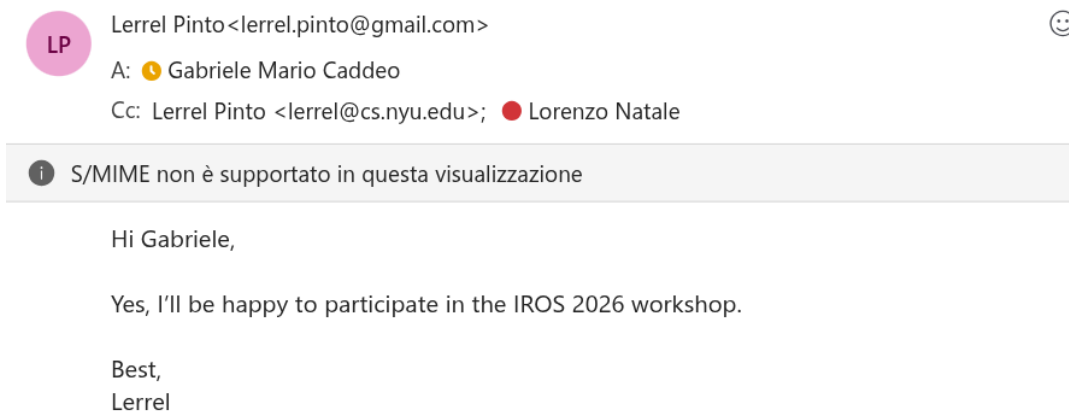


Figure 6: **Prof. Lerrel Pinto** confirmation email.



## Appendix - Confirmation emails from industries

Figure 7: **Alexander Schmitz** confirmation email.

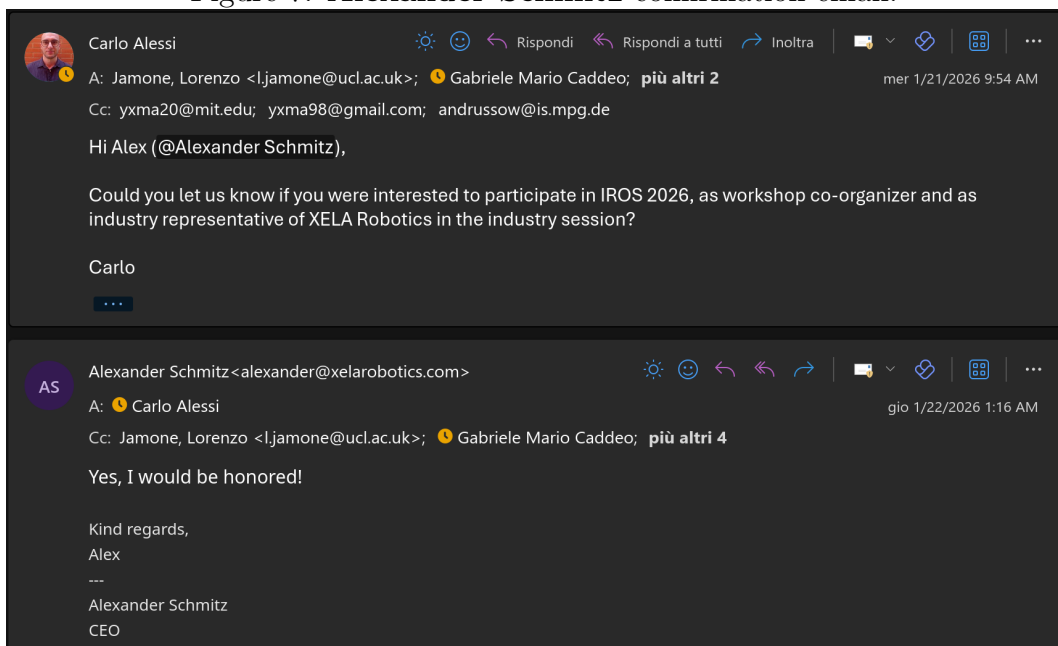


Figure 8: **Matei Ciocarlie** (Tangent Robotics) confirmation.

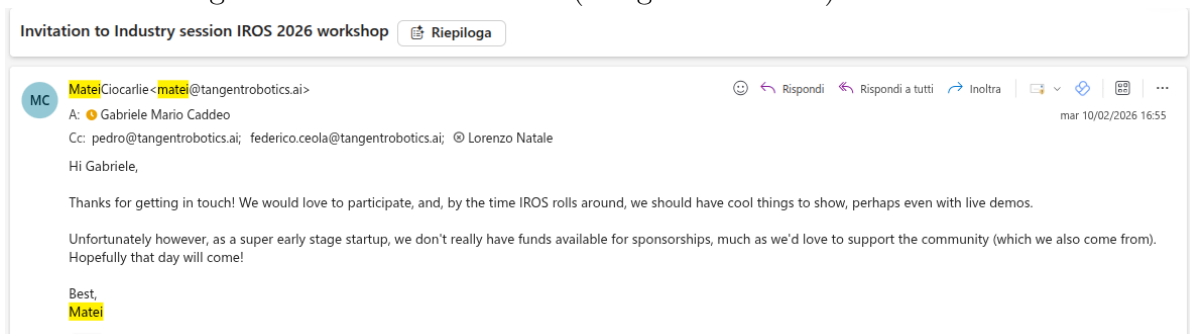


Figure 9: Jeremy Fishel confirmation.

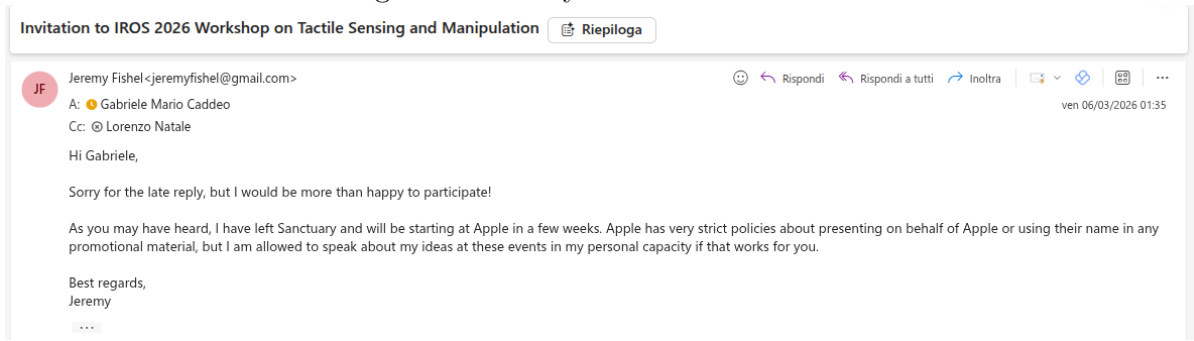


Figure 10: Daimon Robotics confirmation emails.

Dear Dr. Ma,

Thank you for the invitation. We are willing to continue supporting this workshop.

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Best regards,  
Jianghua DUAN / Founder & CEO  
**Daimon Robotics**  
Web: <http://www.dmrobot.com/en/>  
E-mail: [jhduan@dmrobot.com](mailto:jhduan@dmrobot.com)  
Add 1: 2/F, Lee King Industrial Building, Kowloon, Hong Kong.  
Add 2: 13/F, Haina Baichuan Headquarters Building, Bao'an District, Shenzhen, China.

Yuxiang Ma <yxma20@mit.edu> 于2026年3月15日(周日) 上午6:04写道:

Dear Dr Duan,

I hope you are doing well. I wanted to share a quick update: our organizing team has decided to **resubmit the workshop proposal to IROS**, adjusting it to a **half-day format** while keeping the overall theme and structure largely the same.

We would be honored to invite you to continue participating in the IROS workshop. As for the support, we are happy with anything, like some money to donate or one of your product as a prize for paper contributors.

Thank you very much, and we look forward to hearing from you.

Best regards,  
Yuxiang Ma